



Ask Anything

Quick Summary

Alan Turing

British mathematician and logicianAlso known as: *Alan Mathison Turing*

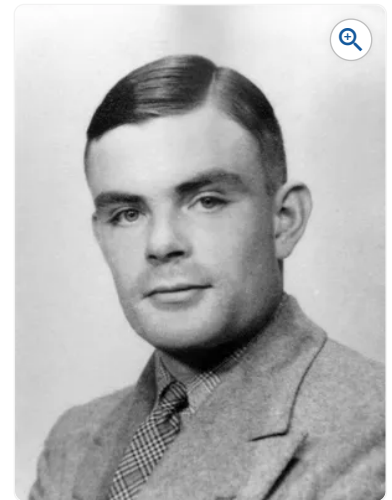
B.J. Copeland Britannica Editors History

★ Britannica AI

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Alan Turing British mathematician Alan Turing, c. 1930s. Turing did the earliest work on AI, and he introduced many of the ...[\(more\)](#)

Alan Turing (born June 23, 1912, [London](#), England—died June 7, 1954, Wilmslow, Cheshire) was a British mathematician and logician who made major contributions to [mathematics](#), [cryptanalysis](#), [logic](#), [philosophy](#), and mathematical [biology](#) and also to the new areas later named [computer science](#), [cognitive science](#), [artificial intelligence](#), and [artificial life](#).

Early life and career

The son of a civil servant, Turing was educated at a top private school. He entered the [University of Cambridge](#) to study mathematics in 1931. After graduating in 1934, he was elected to a fellowship at [King's College](#) (his college since 1931) in recognition of his research in [probability theory](#). In 1936 Turing's [seminal](#) paper "On Computable Numbers, with an Application to the *Entscheidungsproblem* [[Decision Problem](#)]" was recommended for publication by the American mathematical logician [Alonzo Church](#), who had himself just published a paper that reached the same conclusion as Turing's, although by a different method. Turing's method (but not so much Church's) had profound significance for the emerging [science](#) of computing. Later that year Turing moved to [Princeton University](#) to study for a Ph.D. in [mathematical logic](#) under Church's direction (completed in 1938).

The *Entscheidungsproblem*

What mathematicians called an "effective" method for solving a problem was simply one that could be carried by a human mathematical clerk working by rote. In Turing's time, those rote-workers were in fact called "computers," and human computers carried out some aspects of the work later [done](#) by electronic computers. The *Entscheidungsproblem* sought an effective method for solving the fundamental mathematical problem of determining exactly which mathematical statements are provable within a given formal mathematical system and which are not. A method for determining this is called a decision method. In 1936 Turing and Church independently showed that, in general, the *Entscheidungsproblem* problem has no resolution, proving that no consistent [formal system](#) of arithmetic has an effective decision method. In fact, Turing and Church showed that even some purely logical systems, considerably weaker than arithmetic, have no effective decision method. This result and others—notably mathematician-logician [Kurt Gödel](#)'s incompleteness results—dashed the hopes, held by some mathematicians, of discovering a formal system that would reduce the whole of mathematics to methods that (human) computers could carry out. It was in the course of his work on the *Entscheidungsproblem* that Turing invented the universal [Turing machine](#), an abstract computing machine that [encapsulates](#) the fundamental logical principles of the [digital computer](#).

The Church-Turing thesis

An important step in Turing's argument about the *Entscheidungsproblem* was the claim, now called the [Church-Turing thesis](#), that everything humanly computable can also be computed by the universal Turing machine. The claim is important because it marks out the limits of human computation. Church in his work used instead the thesis that all human-computable functions are identical to what he called lambda-definable functions (functions on the positive integers whose values can be calculated by a process of repeated substitution). Turing showed in 1936 that Church's thesis was equivalent to his own, by proving that every lambda-definable function is computable by the universal Turing machine and [vice versa](#). In a review of Turing's work, Church acknowledged the superiority of Turing's formulation of the thesis over his own (which made no reference to computing

machinery), saying that the concept of computability by a Turing machine “has the advantage of making the identification with effectiveness...evident immediately.”

Code breaker

Having returned from the [United States](#) to his fellowship at King’s College in the summer of 1938, Turing went on to join the Government Code and Cypher School, and, at the outbreak of war with Germany in September 1939, he moved to the organization’s wartime headquarters at [Bletchley Park](#), Buckinghamshire. A few weeks previously, the Polish government had given Britain and France details of the Polish successes against [Enigma](#), the principal [cipher](#) machine used by the German military to encrypt radio communications. As early as 1932, a small team of Polish mathematician-cryptanalysts, led by [Marian Rejewski](#), had succeeded in deducing the internal wiring of [Enigma](#), and by 1938 Rejewski’s team had devised a code-breaking machine they called the [Bomba](#) (the Polish word for a type of ice cream). The Bomba depended for its success on German operating procedures, and a change in those procedures in May 1940 rendered the Bomba useless. During the autumn of 1939 and the spring of 1940, Turing and others designed a related, but very different, code-breaking machine known as the [Bombe](#). For the rest of the war, Bombes supplied the Allies with large quantities of [military intelligence](#). By early 1942 the cryptanalysts at Bletchley Park were [decoding](#) about 39,000 intercepted messages each month, a figure that rose subsequently to more than 84,000 per month—two messages every minute, day and night. In 1942 Turing also devised the first systematic method for breaking messages encrypted by the sophisticated German cipher machine that the British called “Tunny.” At the end of the war, Turing was made an Officer of [the Most Excellent Order of the British Empire](#) (OBE) for his code-breaking work.

Quick Facts

In full: Alan Mathison Turing

Born: June 23, 1912, [London, England](#)

Died: June 7, 1954, Wilmslow, Cheshire (aged 41)

Subjects Of Study: [morphogenesis](#) • [artificial intelligence](#) • [computer program](#) • [computability](#) • [...\(Show more\)](#)

Role In: [World War II](#)

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Enigma machine explained World War II saw wide use of codes and ciphers, from substitution ciphers to the wc...(more)

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Pop Quiz: 17 Things to Know About World War II

Computer designer

In 1945, the war over, Turing was recruited to the National Physical Laboratory (NPL) in [London](#) to create an electronic [computer](#). His design for the [Automatic Computing Engine](#) (ACE) was the first complete specification of an electronic stored-program all-purpose [digital computer](#). Had Turing’s ACE been built as he planned, it would have had vastly more memory than any of the other early computers, as well as being faster. However, his colleagues at NPL thought the engineering too difficult to attempt, and a much smaller machine was built, the Pilot Model ACE (1950).

NPL lost the race to build the world’s first working electronic stored-program digital computer—an honour that went to the Royal Society Computing Machine Laboratory at the [University of Manchester](#) in June 1948. Discouraged by the [delays](#) at NPL, Turing took up the deputy directorship of the Computing Machine Laboratory in that year (there was no director). His earlier theoretical concept of a universal [Turing machine](#) had been a fundamental influence on the [Manchester](#) computer project from the beginning. After Turing’s arrival at Manchester, his main contributions to the computer’s [development](#) were to design an input-output system—using [Bletchley Park](#) technology—and to design its programming system. He also wrote the first-ever programming manual, and his programming system was used in the Ferranti [Mark I](#), the first marketable electronic digital computer (1951).

Artificial intelligence pioneer

Turing was a founding father of artificial intelligence and of modern [cognitive science](#), and he was a leading early exponent of the [hypothesis](#) that the human [brain](#) is in large part a digital computing machine. He theorized that the [cortex](#) at birth is an “unorganised machine” that through “training” becomes organized “into a universal machine or something like it.” Turing proposed what subsequently became known as the [Turing test](#) as a [criterion](#) for whether an artificial computer is thinking

(1950). In late 2022, the advent of [ChatGPT](#) reignited conversation about the likelihood that the components of the Turing test had been met.

Last years

Turing was elected a fellow of the [Royal Society](#) of London in March 1951, a high honour, yet his life was about to become very hard. In March 1952 he was convicted of “gross indecency”—that is to say, homosexuality, a crime in Britain at that time—and he was sentenced to 12 months of [hormone](#) “therapy.” Now with a criminal record, he would never again be able to work for Government Communications Headquarters (GCHQ), the British government’s postwar code-breaking centre.

Turing spent the remainder of his short career at Manchester, where he was appointed to a specially created readership in the theory of computing in May 1953. From 1951 Turing had been working on what is now known as [artificial life](#). He published “The Chemical Basis of Morphogenesis” in 1952, describing aspects of his research on the development of form and pattern in living organisms. Turing used Manchester’s Ferranti Mark I computer to model his hypothesized chemical mechanism for the generation of anatomical structure in animals and plants.

In the midst of this groundbreaking work, Turing was discovered dead in his bed, [poisoned by cyanide](#). The official verdict was [suicide](#), but no motive was established at the 1954 inquest. His death is often attributed to the hormone “treatment” he received at the hands of the authorities following his trial for being gay. Yet he died more than a year after the hormone doses had ended, and, in any case, the [resilient](#) Turing had borne that cruel treatment with what his close friend Peter Hilton called “amused fortitude.” Also, to judge by the records of the inquest, no evidence at all was presented to indicate that Turing intended to take his own life, nor that the balance of his [mind](#) was disturbed (as the coroner claimed). In fact, his mental state appears to have been unremarkable at the time. Although suicide cannot be ruled out, it is also possible that his death was simply an accident, the result of his inhaling [cyanide](#) fumes from an experiment in the tiny laboratory adjoining his bedroom. Nor can [murder](#) by the secret services be entirely ruled out, given that Turing knew so much about cryptanalysis at a time when homosexuals were regarded as threats to national security.

By the early 21st century Turing’s [prosecution](#) for being gay had become infamous. In 2009 British Prime Minister [Gordon Brown](#), speaking on behalf of the British government, publicly apologized for Turing’s “utterly unfair” treatment. Four years later Queen [Elizabeth II](#) granted Turing a royal [pardon](#).

B.J. Copeland

Science > Mathematics

 CITE 

computer science

 Geneva G. Belford • All  Britannica Editors  May 25, 2026 • History

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Top Questions

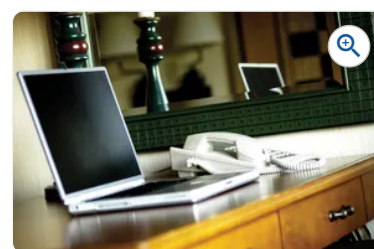
What is computer science?

Who are the most well-known computer scientists?

What can you do with computer science?

Is computer science used in video games?

 Show more



A laptop computer Programmers may use computer science to design portable computers such as laptops.

computer science, the study of [computers](#) and computing, including their theoretical and algorithmic foundations, [hardware](#) and [software](#), and their uses for processing information. The [discipline](#) of computer science includes the study of [algorithms](#) and data structures, computer and [network](#) design, modeling data and information processes, and [artificial intelligence](#).

Computer science draws some of its foundations from [mathematics](#) and [engineering](#) and therefore incorporates techniques from areas such as queueing theory, [probability and statistics](#), and [electronic](#) circuit design. Computer science also makes heavy

use of [hypothesis testing](#) and experimentation during the conceptualization, design, measurement, and refinement of new [algorithms](#), information structures, and computer architectures.

What do you think?

- [Is the Internet "Making Us Stupid?"](#)

Explore the ProCon debate

Computer science is considered as part of a family of five separate yet interrelated disciplines: computer engineering, computer science, [information systems](#), information [technology](#), and software engineering. This family has come to be known collectively as the discipline of computing. These five [disciplines](#) are interrelated in the sense that computing is their object of study, but they are separate since each has its own research perspective and curricular focus. (Since 1991 the Association for Computing Machinery [ACM], the IEEE Computer Society [IEEE-CS], and the Association for Information Systems [AIS] have [collaborated](#) to develop and update the [taxonomy](#) of these five interrelated disciplines and the guidelines that educational institutions worldwide use for their undergraduate, graduate, and research programs.)

The major subfields of computer science include the traditional study of [computer architecture](#), [programming languages](#), and software development. However, they also include computational [science](#) (the use of algorithmic techniques for modeling scientific data), graphics and visualization, human-computer interaction, [databases](#) and information systems, networks, and the social and professional issues that are unique to the practice of computer science. As may be evident, some of these subfields overlap in their activities with other modern fields, such as [bioinformatics](#) and computational [chemistry](#). These overlaps are the [consequence](#) of a tendency among computer scientists to recognize and act upon their field's many interdisciplinary connections.

Development of computer science

[Computer](#) science emerged as an independent discipline in the early 1960s, although the electronic [digital computer](#) that is the object of its study was invented some two decades earlier. The roots of computer science lie primarily in the related fields of [mathematics](#), electrical engineering, [physics](#), and management information systems.

[Mathematics](#) is the source of two key concepts in the development of the computer—the idea that all information can be represented as sequences of zeros and ones and the abstract notion of a "[stored program](#)." In the [binary number system](#), numbers are represented by a sequence of the [binary](#) digits 0 and 1 in the same way that numbers in the familiar [decimal](#) [system](#) are represented using the digits 0 through 9. The relative ease with which two states (e.g., high and low voltage) can be



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[Computers and Technology Quiz](#)

[Electrical engineering](#) provides the basics of circuit design—namely, the idea that electrical impulses input to a circuit can be combined using [Boolean algebra](#) to produce arbitrary outputs. (The Boolean algebra developed in the 19th century supplied a [formalism](#) for designing a circuit with binary input values of zeros and ones [false or true, respectively, in the [terminology](#) of logic] to yield any desired combination of zeros and ones as output.) The [invention](#) of the [transistor](#) and the miniaturization of circuits, along with the invention of electronic, magnetic, and optical media for the storage and transmission of information, resulted from advances in electrical engineering and physics.

[Management information systems](#), originally called [data processing](#) systems, provided early ideas from which various computer science concepts such as sorting, searching, [databases](#), [information retrieval](#), and [graphical user interfaces](#) evolved. Large corporations housed computers that stored information that was central to the activities of running a business—payroll, accounting, inventory management, production control, shipping, and receiving.



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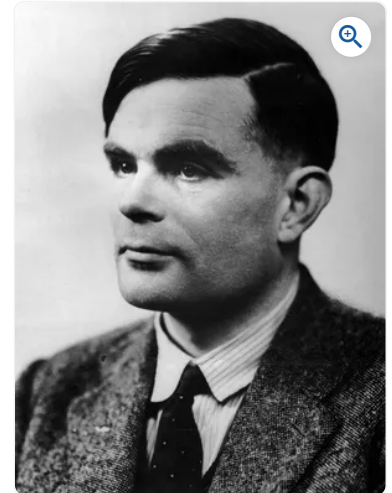
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Theoretical work on computability, which began in the 1930s, provided the needed [extension](#) of these advances to the design of whole machines; a milestone was the 1936 specification of the [Turing machine](#) (a theoretical computational model that carries out instructions represented as a series of zeros and ones) by the British mathematician [Alan Turing](#) and his [proof](#) of the model's computational power. Another breakthrough was the concept of the [stored-program](#) computer, usually credited to Hungarian American mathematician [John von Neumann](#). These are the origins of the computer science field that later became known as [architecture](#) and organization.

In the 1950s, most computer users worked either in scientific research labs or in large corporations. The former [group](#) used computers to help them make complex mathematical calculations (e.g., missile trajectories), while the latter group used computers to manage large amounts of [corporate](#) data (e.g., payrolls and inventories). Both groups quickly learned that writing programs in the [machine language](#) of zeros and ones was not practical or reliable. This discovery led to the development of [assembly language](#) in the early 1950s, which allows programmers to use symbols for instructions (e.g., ADD for addition) and variables (e.g., X). Another program, known as an [assembler](#), translated these symbolic programs into an equivalent binary program whose steps the computer could carry out, or “execute.”



Alan Turing British mathematician Alan Turing conceptualized the Turing machine.

Other [system software](#) elements known as linking loaders were developed to combine pieces of assembled [code](#) and load them into the computer's memory, where they could be executed. The [concept](#) of linking separate pieces of code was important, since it allowed “libraries” of programs for carrying out common tasks to be reused. This was a first step in the development of the computer science field called [software](#) engineering.

Later in the 1950s, assembly language was found to be so cumbersome that the development of high-level languages (closer to natural languages) began to support easier, faster programming. [FORTRAN](#) emerged as the main high-level language for scientific programming, while [COBOL](#) became the main language for business programming. These languages carried with them the need for different software, called [compilers](#), that translate high-level language programs into machine code. As programming languages became more powerful and abstract, building compilers that create high-quality machine code and that are efficient in terms of execution speed and storage [consumption](#) became a challenging computer science problem. The design and implementation of high-level languages is at the heart of the computer science field called programming languages.

Increasing use of computers in the early 1960s provided the [impetus](#) for the development of the first [operating systems](#), which consisted of system-resident software that automatically handled input and output and the execution of programs called “jobs.” The demand for better computational techniques led to a resurgence of interest in numerical methods and their [analysis](#), an activity that expanded so widely that it became known as computational science.

The 1970s and '80s saw the emergence of powerful [computer graphics](#) devices, both for [scientific modeling](#) and other visual activities. (Computerized graphical devices were introduced in the early 1950s with the display of crude images on paper plots and [cathode-ray tube](#) [CRT] screens.) Expensive hardware and the limited availability of software kept the field from growing until the early 1980s, when the [computer memory](#) required for [bitmap graphics](#) (in which an image is made up of small rectangular pixels) became more affordable. Bitmap technology, together with high-resolution display screens and the development of graphics standards that make software less machine-dependent, has led to the explosive growth of the field. Support for all these activities evolved into the field of computer science known as graphics and visual computing.

Closely related to this field is the design and analysis of systems that interact directly with users who are carrying out various computational tasks. These systems came into wide use during the 1980s and '90s, when line-edited interactions with users were replaced by [graphical user interfaces](#) (GUIs). GUI design, which was pioneered by [Xerox](#) and was later picked up by [Apple](#) (Macintosh) and finally by [Microsoft](#) ([Windows](#)), is

important because it [constitutes](#) what people see and do when they interact with a computing device. The design of appropriate user interfaces for all types of users has evolved into the computer science field known as human-computer interaction (HCI).

The field of computer architecture and organization has also evolved dramatically since the first stored-program computers were developed in the 1950s. So called time-sharing systems emerged in the 1960s to allow several users to run programs at the same time from different terminals that were hard-wired to the computer. The 1970s saw the development of the first wide-area [computer networks \(WANs\)](#) and [protocols](#) for transferring information at high speeds between computers separated by large distances. As these activities evolved, they coalesced into the computer science field called networking and communications. A major accomplishment of this field was the development of the [Internet](#).

The idea that instructions, as well as data, could be stored in a computer's memory was critical to fundamental discoveries about the theoretical behavior of [algorithms](#). That is, questions such as, "What can/cannot be computed?" have been formally addressed using these abstract ideas. These discoveries were the origin of the computer science field known as algorithms and complexity. A key part of this field is the study and application of [data](#) structures that are appropriate to different applications. [Data structures](#), along with the development of optimal algorithms for inserting, deleting, and locating data in such structures, are a major concern of computer scientists because they are so heavily used in computer software, most notably in compilers, operating systems, file systems, and [search engines](#).

In the 1960s the invention of [magnetic disk storage](#) provided rapid access to data located at an arbitrary place on the disk. This invention led not only to more cleverly designed file systems but also to the development of [database](#) and information retrieval systems, which later became essential for storing, retrieving, and [transmitting](#) large amounts and wide varieties of data across the Internet. This field of computer science is known as information management.

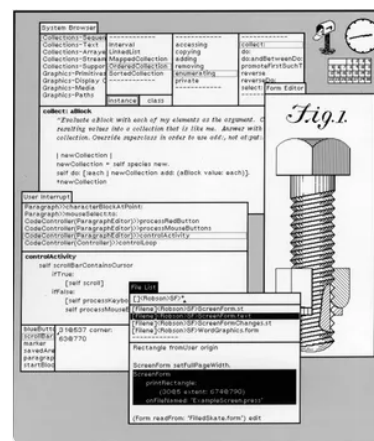
Another long-term goal of computer science research is the creation of computing machines and [robotic](#) devices that can carry out tasks that are typically thought of as requiring [human intelligence](#). Such tasks include moving, seeing, hearing, speaking, understanding natural language, thinking, and even exhibiting [human](#) emotions. The computer science field of intelligent systems, originally known as [artificial intelligence](#) (AI), actually predates the first [electronic](#) computers in the 1940s, although the term *artificial intelligence* was not coined until 1956.

Three developments in computing in the early part of the 21st century—mobile computing, [client-server computing](#), and computer hacking—contributed to the emergence of three new fields in computer science: platform-based development, parallel and [distributed computing](#), and security and information [assurance](#). Platform-based development is the study of the special needs of mobile devices, their operating systems, and their applications. Parallel and distributed computing concerns the development of architectures and programming languages that support the development of algorithms whose components can run simultaneously and asynchronously (rather than sequentially), in order to make better use of time and space. Security and information assurance deals with the design of computing systems and software that protects the [integrity](#) and security of data, as well as the privacy of individuals who are characterized by that data.

Finally, a particular concern of computer science throughout its history is the unique societal impact that accompanies computer science research and technological advancements. With the emergence of the Internet in the 1980s, for example, software developers needed to address important issues related to information security, personal privacy, and system reliability. In addition, the question of whether computer software constitutes [intellectual](#) property and the related question "Who owns it?" gave rise to a whole new legal area of licensing and licensing standards that applied to software and related [artifacts](#). These concerns and others form the basis of social and professional issues of computer science, and they appear in almost all the other fields identified above.

So, to summarize, the discipline of computer science has evolved into the following 15 distinct fields:

- Algorithms and complexity
- Architecture and organization
- Computational science



Graphical user interface The Xerox Alto was the first computer to use graphical icons and a mouse to control the system—th...(more)

- Graphics and visual computing
- Human-computer interaction
- Information management
- Intelligent systems
- Networking and communication
- Operating systems
- Parallel and distributed computing
- Platform-based development
- [Programming](#) languages
- Security and information assurance
- Software engineering
- Social and professional issues

Related Topics: [cryptology](#) • [List of Influential Women and Nonbinary People in Computing](#) • [More...](#)*(Show more)*

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Computer science continues to have strong mathematical and engineering roots. Computer science bachelor's, master's, and doctoral degree programs are routinely offered by postsecondary academic institutions, and these programs require students to complete appropriate mathematics and engineering courses, depending on their area of focus. For example, all undergraduate computer science majors must study discrete mathematics (logic, [combinatorics](#), and elementary [graph theory](#)). Many programs also require students to complete courses in [calculus](#), [statistics](#), [numerical analysis](#), physics, and principles of engineering early in their studies.